JUE WANG

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EDUCATION

University of Toronto — BASc in Computer Engineering Sept 2023 – Apr 2027
Edward S. Rogers Sr. Department of Electrical and Computer Engineering *Toronto, Canada*

Cumulative GPA: 3.58 / 4.00; Major (CE) GPA: 3.72 / 4.00; Expected graduation: Apr 2027

RESEARCH INTERESTS

Embodied AI; Vision-Language Models; Active Perception & Spatial Reasoning; Uncertainty-Aware Robot Learning.

PUBLICATIONS & PREPRINTS

1. **Wang, J.** *"PsyPointer: A Physically Measured Ordinal Feedback Model of VLMs for Closed-Loop Language-Guided Pointing."* Under review, IEEE International Conference on Robotics and Automation (ICRA) 2027.
2. **Wang, J.** *"CalibProbe: Evaluating Active Spatial Belief Repair with Physically Calibrated VLM Error Models."* Under review, CoRL 2026 Workshop on Agentic Robotics.

RESEARCH EXPERIENCE

Independent Research — VLM-Space-Pointer: Embodied VLM Active Perception Jun 2026 – Present
Independent research program | Code: github.com/joewang0430/vlm-space-pointer *Toronto, Canada*

- Built a US$180 actuator-space psychophysics instrument — webcam, pan servo, laser pointer, Arduino, and a camera–laser co-alignment rig — that commands known offsets from an anchored target and so manufactures its own physical ground truth for language-guided pointing.
- Measured a physically grounded ordinal feedback model of four VLMs from 2,304 scored queries — direction-error curve, hit-report channel, and just-noticeable difference (2.8°–7.1° for GPT-4o) — and proved a budget-limited minimax accuracy bound at that measured scale.
- Drove a probabilistic-bisection controller with the measured model: across 383 pre-registered hardware trials on 8 objects in two scenes, ordinal closed-loop correction raised hit rate from 27% for one-shot coordinate regression to 77% (first-author paper under review, IEEE ICRA 2027).
- Extended the calibrated error models into CalibProbe, a hardware-calibrated simulation testbed for active spatial belief repair, where expected-information-gain probing under matched query budgets cut GPT-5.5's gross-error rate from 16% to 0.05% within five probes (first-author paper under review, CoRL 2026 Workshop on Agentic Robotics).

Research Assistant — Intelligent Course Q&A System May 2026 – Present
University of Toronto ECE | Supervisor: Prof. Salma Emara *Toronto, Canada*

- Built Quiz Mode, an LLM-driven adaptive assessment feature that generates course-grounded quizzes, grades answers in real time, and adapts follow-up question difficulty to student performance; validated via structured manual evaluations (100% grading accuracy across 24 cases, 8 adaptive test threads).
- Redesigned chat data structure and Aerospike schema to support multiple concurrent conversations per course.

Research Intern — LLM-Based Government Planning Generation Jun 2025 – Aug 2025
Zhejiang Lab | Details: www.wangjue.me/projects/planning_agent *Hangzhou, China*

- Developed an end-to-end retrieval-augmented generation (RAG) system for drafting government planning documents, using ChromaDB, LangChain, and external knowledge bases to ground long-form output.
- Designed AI-agent workflows for section-wise generation and paragraph-level revision, prepared technical documentation, and filed a software copyright application for the completed system.

INDUSTRY EXPERIENCE

DingTalk, Alibaba Group — Software Engineering Intern Jun 2024 – Aug 2024
Hangzhou, China

- Automated performance testing with the Hypium API in Python, measuring page-load time and frame rate across 11 application sections and multiple interaction scenarios.
- Implemented a HarmonyOS chat-list long-press menu in ArkTS and built crash-analysis tooling that automatically parsed device logs to localize failure points.

RESEARCH & ENGINEERING PROJECTS

FPGA-Assisted Token-Level Inference for a Tiny Decoder-Only LLM 2026 – Present
Independent Research & Engineering Project

- Architecting an ARM-FPGA SoC hardware-software co-design on the AMD Kria KV260 platform, offloading token-level Transformer inference (attention, KV-cache, MLP, greedy decoding) to the programmable logic while using the on-board Arm Cortex-A53 for text tokenization and AXI-bus runtime control.
- Developing a PyTorch software baseline and implementing INT4 GEMV / Transformer-block hardware acceleration, actively profiling the system to optimize latency-per-token, throughput, and FPGA resource utilization (LUT/DSP/BRAM).

ReverC — Browser-Based Reversi Competitive Platform Jun 2025 – Present
Independently developed project | *Website: www.reverc.org*

- Independently built and deployed a sandboxed code-competition platform with LLM-integrated gameplay; adopted as a practice platform for APS105, a first-year programming course at the University of Toronto (Winter 2026), hosting 2,500+ games.

SpeakEasy — AI-Powered Interview Practice Tool Sep 2025
Hackathon project | *Details: devpost.com/software/speakeasy-vdkc9n*

- Engineered a real-time behavioral feedback system by extracting geometric features via MediaPipe joint parsing; the project was named a Hack the North 2025 finalist (12 of 250+ teams).

TECHNICAL SKILLS

AI / ML / Research: PyTorch, Ollama, LangChain, ChromaDB, MediaPipe, LLM APIs; Bayesian inference, probabilistic modeling, psychometrics, experimental design.

Programming: Python, C/C++, Verilog, SQL (PostgreSQL, SQLite), JavaScript/TypeScript, RISC-V/x86 Assembly, C#.

Systems / Embedded / Web: Git, Node.js, Django, WebSockets, ESP32, Arduino, Next.js, FastAPI.

HONORS & AWARDS

- Ranked among top 30 of ~350 ECE students by academic standing (2024).
- Three-time Dean's Honour List (Fall 2023, Winter 2024, Fall 2025) — University of Toronto.
- Edward S. Rogers Sr. Scholarship — University of Toronto.
- Hack the North 2025 Finalist — SpeakEasy; one of 12 finalist teams among 250+ teams / 1,000+ hackers.
- GenAI Genesis 2024, Best AI in Equity, Diversity & Inclusion — SafetyBud; an AI-powered safety training app.

RELEVANT COURSEWORK

AI & Machine Learning	APS360 Applied Fundamentals of Deep Learning [89, A] CSC384 Introduction to Artificial Intelligence ECE421 Introduction to Machine Learning [In Progress] CSC401 Natural Language Computing [In Progress]
Systems & Algorithms	ECE345 Algorithms and Data Structures [91, A+] ECE344 Operating Systems ECE243 Computer Organization ECE552 Computer Architecture [In Progress]
Mathematical Foundations	MAT188 Linear Algebra [92, A+] MAT186 Calculus I [88, A]; MAT187 Calculus II [94, A+] ECE302 Probability and Applications